Post-YOLO action selection

Processing, numerical performance, reference labels and efficiency | October 29, 2026

Recommendation: use General Algorithmic Approach for the predefined robot plans. Use A* or UCS when the goal is an economical sequence of object reviews. The current LLM pipeline needs reliability and loading improvements before use for responsive decisions.

Recommended order for efficient, usable actions

This is an engineering recommendation for the current demo, not a universal performance score. It weighs policy fit and usable output, review-plan cost, and computation. Ties avoid overstating small timing differences. For a single review choice, probability inference can be preferred to search; if only calculation speed matters, DFS moves up.

Rank	Method	Why it is placed here	Median response
1	General robot plans	Most direct match for the demo robot policy; selects a named plan and preserves all matching action steps.	0.337 ms
2	A*	Best tier for economical review ordering; optimal on all 100 small graphs.	0.272 ms
2	UCS	Same optimal-plan result as A*; more work in the 1,000-target stress test.	0.279 ms
3	Probability enumeration	One review recommendation without graph expansion; also estimates the scene.	0.533 ms
4	Greedy best-first	Quick approximate ordering; lower compute cost but no minimum-cost guarantee.	0.279 ms
5	BFS	Complete plans, but shortest depth does not minimize weighted review effort.	0.266 ms
5	IDS	Same plan quality as BFS here, with repeated exploration and slower scaling.	0.293 ms
6	DFS	Fast plan generation; its selected review plans had the largest cost overhead.	0.267 ms
7	Qwen3-VL 2B	Rich contextual explanation, but slow and frequently rejected by current validation.	10.24 s

Observed responses on the 24 public images

Robot plans: 21 Take Action and 3 No Action. Its decisions matched 20/24 annotation-derived robot decisions (83.3%); exact plan names matched only 10/24. All 24 results agreed with the robot rules applied to the actual YOLO detections.

Each search and probability: 14 Take Action and 10 No Action. All matched the shared review rules on the actual detections in 24/24 cases; annotation-derived review agreement was 15/24 (62.5%). Identical binary decisions did not mean identical targets or review order.

LLM: 1 validated Take Action, 11 validated No Action, 11 rejected/fallback responses, and 1 no-detection bypass. The interface displayed No Action for the rejected responses, but these were **not usable decisions**. Across both repetitions, 22/46 actual inference runs fell back. After validation, first-pass agreement was 9/24 with review references and 2/24 with robot references.

Accuracy qualification: COCO labels objects, not required robot actions. Scores measure agreement with two explicit policies, not independently human-labeled action correctness. The robot reference has 21 positives and 3 negatives: always acting scores 87.5%, versus robot plans at 83.3%; balanced accuracy is 50% versus 61.9%. Review references request signal/tool review; robot references also react to people, animals, vehicles and counts. Predicted scores below 50% trigger verification, while annotations are treated as confirmed. The recommendation concerns policy fit and efficiency, not proven real-world safety.

Numerical decision performance

Same 24 images for every method | Positive class: Take Action | Values are percentages

Reference truth: COCO supplies human object labels/boxes. A fixed review policy or robot policy converts those annotations into expected binary actions. These are **policy-derived action references**, not human judgments about what a robot should do. Every method is scored against both references, using repetition 0 only.

Robot-policy reference: 21 Action / 3 No Action

Method	Accuracy	Precision	Recall	F1	Balanced	Available
General robot plans	83.3	90.5	90.5	90.5	61.9	100.0
A*	54.2	85.7	57.1	68.6	45.2	100.0
UCS	54.2	85.7	57.1	68.6	45.2	100.0
Probability enumeration	54.2	85.7	57.1	68.6	45.2	100.0
Greedy best-first	54.2	85.7	57.1	68.6	45.2	100.0
BFS	54.2	85.7	57.1	68.6	45.2	100.0
IDS	54.2	85.7	57.1	68.6	45.2	100.0
DFS	54.2	85.7	57.1	68.6	45.2	100.0
Qwen3-VL 2B	8.3	100.0	4.8	9.1	19.0	54.2

Review-policy reference: 7 Action / 17 No Action

Method	Accuracy	Precision	Recall	F1	Balanced	Available
General robot plans	33.3	28.6	85.7	42.9	48.7	100.0
A*	62.5	42.9	85.7	57.1	69.3	100.0
UCS	62.5	42.9	85.7	57.1	69.3	100.0
Probability enumeration	62.5	42.9	85.7	57.1	69.3	100.0
Greedy best-first	62.5	42.9	85.7	57.1	69.3	100.0
BFS	62.5	42.9	85.7	57.1	69.3	100.0
IDS	62.5	42.9	85.7	57.1	69.3	100.0
DFS	62.5	42.9	85.7	57.1	69.3	100.0
Qwen3-VL 2B	37.5	0.0	0.0	0.0	26.5	54.2

Reading the metrics: Accuracy is correct outputs / 24; precision is TP/(TP+FP); recall is TP/(TP+FN+AP); F1 is 2TP/(2TP+FP+FN+AP). Balanced accuracy averages positive recall and negative recall TN/(TN+FP+AN). Available is the share of usable decisions, including the no-detection bypass. AP/AN are unavailable outputs on positive/negative references. They stay in denominators and never count as correct No Action. Confusion counts follow on the next page.

Baselines: Always acting achieves robot accuracy 87.5%, F1 93.3%, balanced accuracy 50.0%. Always choosing No Action achieves review accuracy 70.8%, F1 0.0%, balanced accuracy 50.0%. Class imbalance makes raw accuracy and positive F1 insufficient on their own.

LLM interpretation: its 100% robot-policy precision comes from just one accepted action (TP=1, FP=0); recall is 4.8%, F1 9.1%, and availability 54.2%. It does not establish strong action recognition. Robot-plan accuracy 20/24 has a 95% Wilson interval of 64.1%-93.3%; all per-method intervals and macro-F1 scores are in performance_metrics.csv.

Latency, work and failure counts

Measured post-YOLO response times | All repetitions for timing; unique images for correctness

Method	Median	95th pct.	Unavailable	1,000-box median	Optimal plans*	Mean cost ratio*
General robot plans	0.337 ms	0.511 ms	0/120	16.17 ms	N/A	N/A
A*	0.272 ms	1.004 ms	0/120	193.04 ms	100/100	1.00x
UCS	0.279 ms	1.566 ms	0/120	215.31 ms	100/100	1.00x
Probability enumeration	0.533 ms	0.774 ms	0/120	16.41 ms	N/A	N/A
Greedy best-first	0.279 ms	0.711 ms	0/120	35.46 ms	31/100	1.10x
BFS	0.266 ms	1.194 ms	0/120	153.35 ms	33/100	1.08x
IDS	0.293 ms	1.667 ms	0/120	279.13 ms	33/100	1.08x
DFS	0.267 ms	0.568 ms	0/120	23.29 ms	0/100	1.49x
Qwen3-VL 2B	10.24 s	12.76 s	22/48	Not tested	N/A	N/A

* Separate synthetic graphs with 1-5 targets. Mean cost ratio = returned cost / independently enumerated optimum; 1.00x is best. Search plans covered every target. The 1,000-box stress uses three repetitions and is not part of public-image accuracy.

Classical latency summarizes 120 calls each; LLM latency summarizes 46 inference calls, excluding 2 empty-detection bypasses. Unavailable counts include all measured calls: 22/48 LLM outputs failed validation, or 22/46 (47.8%) of actual inference calls. First calls are excluded from these timing distributions and saved separately. Latency includes input preparation and explanation, not just the decision step.

Confusion counts against the robot-policy reference

Method	TP	TN	FP	FN	AP	AN
General robot plans	19	1	2	2	0	0
A*	12	1	2	9	0	0
UCS	12	1	2	9	0	0
Probability enumeration	12	1	2	9	0	0
Greedy best-first	12	1	2	9	0	0
BFS	12	1	2	9	0	0
IDS	12	1	2	9	0	0
DFS	12	1	2	9	0	0
Qwen3-VL 2B	1	1	0	11	9	2

TP/TN: correct accepted Action/No Action. FP: accepted Action on a No Action reference. FN: accepted No Action on an Action reference. AP/AN: no usable result on Action/No Action references. Every row sums to 24. Rejected LLM outputs are shown separately rather than silently treated as predictions. Review-policy counts are in the accompanying CSV/JSON.

Resource context: robot plans always check 12 rules / 18 conditions; probability enumerates 20 scene hypotheses. LLM medians were 1,285 input / 67 output tokens, with about 3.1 GiB model residency in one snapshot and 6.40 s reported loading time per inference. These work units are not interchangeable. No energy or peak-memory comparison was measured. All eight classical approaches passed 423 independent policy cases each.

What each algorithm actually did

Shared input: YOLO filters at confidence 0.25; valid boxes are normalized, sorted by score and assigned d1, d2, etc. Classical methods process every eligible box. The LLM also receives image pixels, with at most 10 boxes and possible further prompt truncation. Each description below explains the next processing step; results use the first pass over 24 images.

General Algorithmic Approach - predefined robot plans

Grouped eligible detections by label and confidence, counted them and evaluated 12 rules / 18 conditions. The first matching action plan took priority over idle plans; other matching action steps were retained. Selected 10 uncertainty, 3 vehicle, 2 people, 2 signal, 2 animal, 1 people-and-vehicles, and 1 crowd action plans; 2 sparse and 1 empty plans requested no action. Output included the plan name, matched rules and recommended steps.

A* Search - economical review sequence

Flagged tentative labels, signals and tools, then ordered all triggered reviews in batches of five using accumulated cost plus a lower-bound heuristic. First suggestions were verification in 12 images, signal checks in 2, and no action in 10. Found the minimum-cost plan on all 100 synthetic graphs. It changed review order, not the rules deciding whether review was needed.

Uniform-Cost Search - cost-first review sequence

Used the same targets and action choices as A*, expanding the lowest accumulated-cost state without a heuristic. Returned the same first-action counts as A* and optimal plans on 100/100 synthetic graphs. In the large stress case it took 215.31 ms versus A* at 193.04 ms; optimality applies within each batch, not across the complete multi-batch plan.

Probability Algorithm - strongest review recommendation

Combined distinct-label likelihoods with each scene prior, normalized 20 scene hypotheses, and applied hand-authored review probabilities to every eligible detection. Chose the highest-probability review above its threshold: 12 verification requests, 2 signal checks and 10 no-action results. It did not search for a review route. Probability values express a teaching policy, not calibrated likelihood that an action is correct.

Greedy Best-First Search - fast approximate ordering

Expanded the state with the smallest estimated remaining review cost, ignoring cost already paid. Returned 12 verification, 2 signal-check and 10 no-action first suggestions. Plans were optimal in 31/100 synthetic graphs, with mean cost 1.10 times optimum. Its 1,000-target response was much faster than A* but did not preserve minimum-cost plans.

Breadth-First Search - shallowest complete plan

Represented a state as reviewed-target mask plus last target, then expanded states in first-in, first-out order until every triggered target was scheduled. Returned 12 verification, 2 signal-check and 10 no-action first suggestions. All complete plans have one step per target, so a shortest-depth solution offers no weighted-cost advantage here. It was optimal in 33/100 synthetic graphs, averaging 1.08 times optimum.

Iterative Deepening Search - repeated depth-first passes

Repeated depth-limited DFS with increasing limits until it found a complete review plan. Its first suggestions and synthetic plan quality matched BFS: 12 verification, 2 signal-check, 10 no-action, and 33/100 optimal plans. Repeated exploration made it the slowest classical approach in the 1,000-target stress test, without a better action plan in these tests.

Depth-First Search - quick first complete plan

Followed a deep branch before trying alternatives. Returned 12 verification, 2 generic inspect suggestions and 10 no-action results. In the two signal cases it selected inspect rather than check_signal. It generated plans quickly, but was optimal on 0/100 synthetic graphs and averaged 1.49 times optimum. That result describes this implementation and fixture, not every possible DFS application.

Qwen3-VL 2B - visual choice plus generated explanation

Read the image and a bounded YOLO evidence list, selected environment/setting/action, then generated an explanation, with one retry when needed. The sole accepted action was to verify a tentative person detection in a skiing image. Three first-stage choices contradicted the allowed environment/setting combination; eight more images failed later explanation validation. The second pass reproduced the same decisions and statuses. Its richer output did not produce dependable action selection here.

All actions above are recommendations. This application sends no physical robot commands.

How they responded to the same scenes

Concrete recorded examples - paraphrases preserve the selected action and target.

Street intersection - COCO image 555050

Reference: review = Take Action; robot = Take Action, primary plan People-and-Vehicles. YOLO returned 12 boxes. Search planned 10 reviews. The search methods agreed that review was needed, but differed in target order and effort. Costs below are the full review-plan cost, not execution time.

Method	First response	Full plan cost
A*	Check actual signal state [d4]	13.165
UCS	Check actual signal state [d4]	13.165
BFS	Check actual signal state [d2]	15.578
IDS	Check actual signal state [d2]	15.578
Greedy best-first	Check actual signal state [d5]	16.653
DFS	Inspect [d2]	17.378

Probability: check signal [d2], without planning an order. **Robot plans:** Signal-Review selected; animal, uncertainty and vehicle steps also retained. The selected Signal-Review plan matched the binary reference but not its primary plan name. **LLM:** returned outdoor + shop + no_action; validation rejected the setting combination, so the application displayed a fallback. A* / UCS cost 13.165 versus DFS at 17.378, about 24% lower in this scene.

Three other useful examples

Scene	Observed behavior
Skiing - 43314	References: review No Action; robot Take Action / People-Presence. A* / UCS reviewed tentative skis first (cost 2.200); BFS, DFS, IDS and greedy reviewed the tentative person first (2.283). Probability and LLM also selected person verification. Robot plans requested another view for both tentative detections.
Dog - 206831	References: review No Action; robot Take Action / Animal-Presence. All search methods, probability and LLM returned No Action. Robot plans selected Animal-Presence and conditionally recommended pausing to review positions if moving nearby. This is a policy difference, not proof one answer is universally correct.
Boats - 503823	References: review No Action; robot Take Action / Vehicle-Presence. YOLO returned no boxes. Classical review methods and LLM returned no object-based action; the LLM was not invoked. Robot plans selected Empty-Area / No Action. None could recover the missing detections.

Scope, timing and limitations

24 COCO val2017 images, three each from eight object-based strata, selected with seed 20261029 before outputs. This is one dataset with varied scenes. YOLO26n ran once per image at 640 pixels, confidence 0.25, FP16, max 100 boxes. Every method reused the frozen detections; only the LLM also read pixels. Full reference provenance is on the last page.

Hardware: Ryzen 9 5900HX and RTX 3050 Ti Laptop GPU, 4 GB. Each classical method had 120 measured calls; LLM had 48, including 2 bypasses excluded from its latency. Agreement counts distinct images from repetition 0. Timings include preparation, inference, validation and explanation; exclude YOLO, client construction, queueing, GUI and saving. First calls were separate; LLM first call was 14.75 s. Its model residency snapshot was about 3.1 GiB, with median reported loading time 6.40 s per inference. Peak memory and energy were not measured.

Synthetic tests were separate: 423 policy cases per classical method passed; 100 graphs independently checked plan cost; 1,000-target timing used three repetitions. Search cost is illustrative effort plus image-coordinate attention shifts, not physical distance. Public-image sample size is small (20/24 has about 64%-93% Wilson interval). Full latency distributions, response counts and concrete outputs accompany the PDF.

Sources: COCO data | Tested implementation: 61c9a7f. Revision changes the report only; measurements are unchanged.

Ground truth and reference labels

Auditable sources and all 24 expected actions

Object ground truth: official COCO val2017 instance annotations (annotation IDs, category IDs and boxes), retaining iscrowd=0. The saved source member is annotations/instances_val2017.json; its SHA-256 and selected annotations are in ground_truth.json. No algorithm output or LLM judgment was used to generate these labels.

Action reference rules: review = Action for any annotated traffic light, stop sign, knife or scissors. Robot = Action for any annotated person, vehicle, signal, animal or tool, or at least 10 annotated objects. Primary-plan priority: crowd (5+ people), people+vehicles, signal, animal, tool, dense objects, vehicle, people; otherwise empty, sparse (1-3), or moderate (4-9). Exact label groups are saved in JSON. All annotations are treated as confirmed; verification of uncertain detections therefore has no annotation-derived counterpart.

COCO image ID	Sample stratum	Objects	Review	Robot	Primary robot reference
000000157365	signals	9	Action	Action	People-and-Vehicles
000000335800	signals	1	Action	Action	Signal-Review
000000555050	signals	18	Action	Action	People-and-Vehicles
000000130613	tools	5	Action	Action	Tool-Handling
000000010583	tools	4	Action	Action	Tool-Handling
000000174123	tools	6	Action	Action	Tool-Handling
000000361180	animals	1	None	Action	Animal-Presence
000000121586	animals	15	None	Action	Animal-Presence
000000206831	animals	1	None	Action	Animal-Presence
000000458702	transport	4	None	Action	Vehicle-Presence
000000156071	transport	16	None	Action	Crowded-Area
000000108440	transport	1	None	Action	Vehicle-Presence
000000014226	kitchen/dining	8	None	Action	People-Presence
000000429623	kitchen/dining	4	None	None	Moderate-Object
000000391722	kitchen/dining	9	None	Action	Crowded-Area
000000529122	home/office	19	Action	Action	Tool-Handling
000000458768	home/office	6	None	None	Moderate-Object
000000007795	home/office	4	None	None	Moderate-Object
000000128748	sports	5	None	Action	People-Presence
000000043314	sports	5	None	Action	People-Presence
000000239773	sports	6	None	Action	People-Presence
000000503823	air/water	4	None	Action	Vehicle-Presence
000000096549	air/water	1	None	Action	Vehicle-Presence
000000336309	air/water	1	None	Action	Vehicle-Presence

Action = Take Action; None = No Action. Counts come from annotations, not YOLO predictions. ground_truth.csv includes the exact counts by label and image URL. Search-plan optimality has a separate numerical reference: exhaustive enumeration of all orders and minimum effort choices in each synthetic graph. Neither reference proves physical navigation safety or correct real-world intervention.

Sources: COCO download | Official COCO API. Measurements remain from source commit 61c9a7f; this revision adds derived metrics and documented references, not new model runs.

Search trees: depth and order

Recorded example: COCO 43314 (skiing), first measured repetition

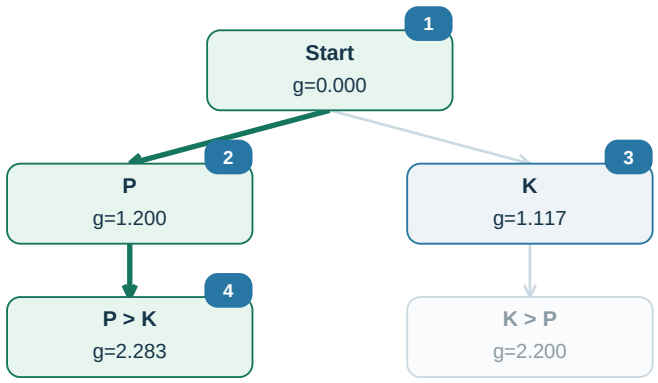
d1 | skis | 58.15%
Above 50% review cutoff

d2 | person | 47.66%
Tentative: verification target

d3 | skis | 25.98%
Tentative: verification target

P = verify person d2; **K** = verify skis d3. d1 needs no review. Each node is a partial review sequence; both leaves cover both targets. Blue badges show actual visit numbers; green shows the returned path. Gray nodes were not visited before stopping.

BFS - breadth-first search

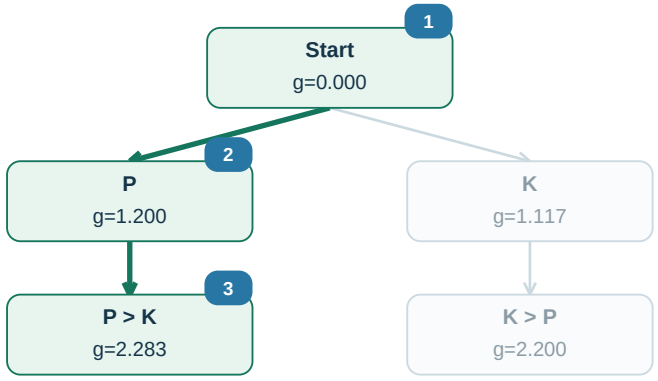


A queue visits all shallower states first. Both one-step choices are visited before the first complete plan. Equal depth does not imply equal cost.

Visit order: Start > P > K > P,K
First recommendation: verify person.

4 visits | 5 generated
Returned cost: **2.283**

DFS - depth-first search

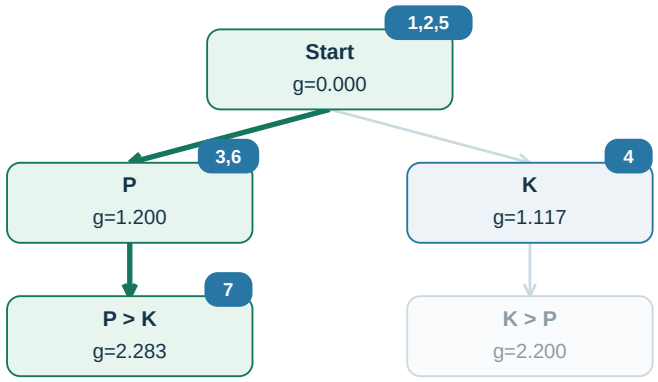


A stack follows the first branch to a complete plan. It stops before visiting K first, so it misses the cheaper complete plan in this example.

Visit order: Start > P > P,K
First recommendation: verify person.

3 visits | 4 generated
Returned cost: **2.283**

IDS - iterative deepening



Depth-limited DFS restarts with limits 0, 1 and 2. Multiple numbers on a node show revisits. It returns the same plan as BFS here.

Limits: 0: Start
1: Start > P > K
2: Start > P > P,K

7 visits | 8 generated
Returned cost: **2.283**

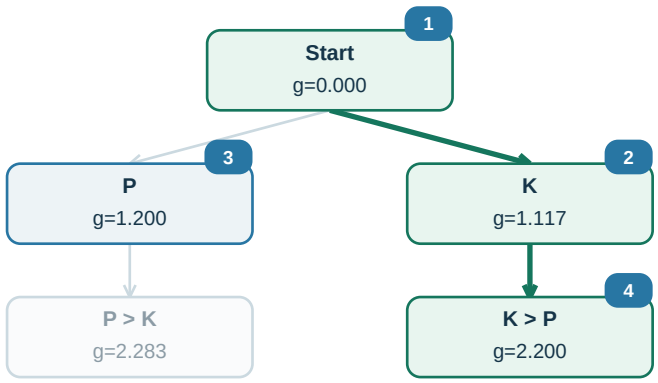
Search trees: cost and heuristics

Same two targets and possible tree | Different rules choose the next state

g is cost already paid; **h** estimates remaining cost; **f = g + h**. A review costs 1 unit plus movement between normalized box centers. Starting at image center, K costs 1.117 and P costs 1.200; moving between them adds a 1.083 review step.

P > K totals 2.283; K > P totals 2.200. At either one-step node, h = 1.083. These are review-effort units, not milliseconds or robot travel distance. Blue badges = visits; green = selected path.

UCS - uniform-cost search

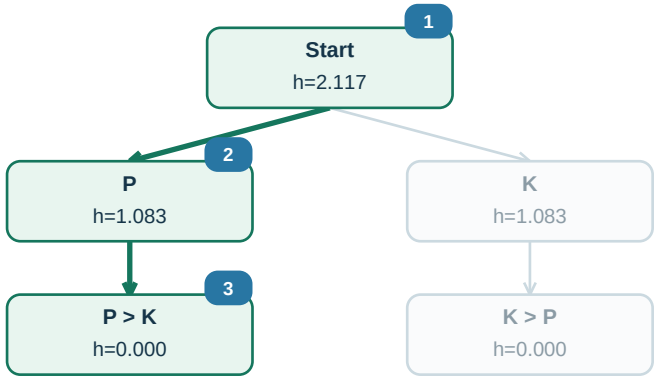


Selects the smallest g. K is cheaper than P, but P is still cheaper than either complete plan, so UCS visits it before returning K > P.

Visit order: Start > K > P > K,P
First recommendation: verify skis.

4 visits | 5 generated
Returned cost: **2.200**

Greedy best-first search

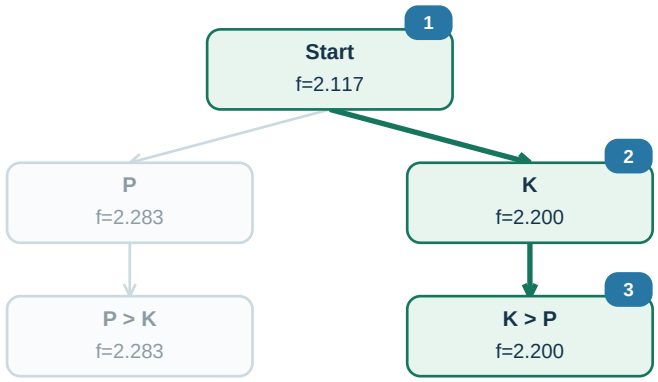


Selects the smallest h and ignores g. P and K tie at h=1.083; insertion order favors P. The next goal has h=0, so it stops.

Visit order: Start > P > P,K
First recommendation: verify person.

3 visits | 4 generated
Returned cost: **2.283**

A* search



Selects the smallest f. K has f=2.200 versus P at 2.283. The K > P goal also has f=2.200, so A* returns it with fewer visits than UCS.

Visit order: Start > K > K,P
First recommendation: verify skis.

3 visits | 4 generated
Returned cost: **2.200**

Probability: two calculations

Recorded skiing example | Scene estimation and action selection use separate calculations

d1 | skis | 58.15%
Above 50% review cutoff

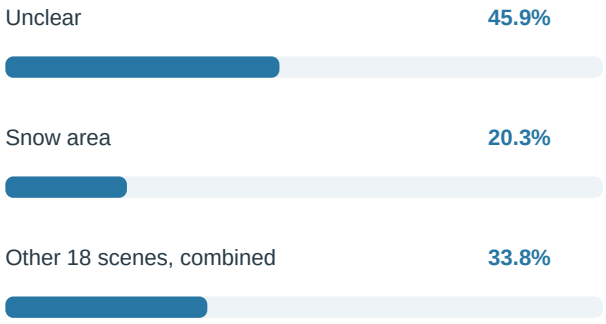
d2 | person | 47.66%
Tentative: verification target

d3 | skis | 25.98%
Tentative: verification target

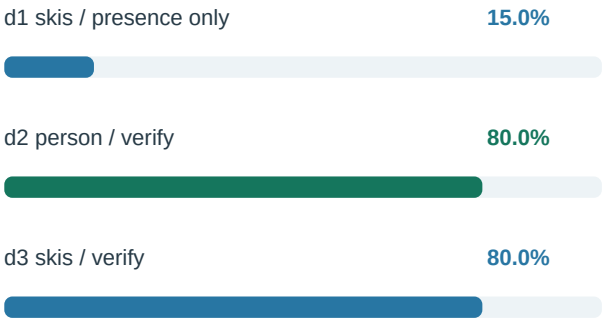
1A. Estimate the scene
Count each label once at its highest score: skis 58.15%, person 47.66%. Combine scene priors and label likelihoods for all 20 hypotheses.

1B. Rate each possible review
Keep all three boxes. Look up P(review useful | label, confidence band) in the hand-authored policy table.

2A. Normalize the scene scores
posterior = prior x likelihoods / sum of scores



2B. Select the largest review score
max(0.15, 0.80, 0.80) = 0.80



3A. Output: unclear
The leading hypothesis is unclear. No specific scene meets the acceptance gate (two strong informative cues, at least 55%, and a 15-point lead).

3B. Output: verify person d2
80% exceeds the 50% threshold. d2 and d3 tie; the first maximum, d2, is chosen. This method selects one review, without ordering a plan.

A concrete scene calculation

Unclear score = 0.30 x 0.18 x 0.55 = **0.02970**. Divide by the sum over 20 scenes, 0.06476, to obtain **45.86%**. The implementation uses log scores for numerical stability. Low-confidence label likelihoods are weakened; detector confidence itself is not the probability of a scene.

The scene estimate does not feed into the review-score table. Both sets of probabilities are hand-authored model values, not measured accuracy. The review reference for this image is No Action; the tentative YOLO labels cause a verification request.

Robot plans: count, match, select

Recorded skiing example | A fixed rule table chooses a named action plan

d1 | skis | 58.15%
Above 50% review cutoff

d2 | person | 47.66%
Tentative: verification target

d3 | skis | 25.98%
Tentative: verification target

1. Build counts from eligible boxes

Total 3 | strong 1 | tentative 2 | strong people 0 | vehicles / signals / animals / tools 0.
Only scores at least 50% enter strong category counts. The detected person at 47.66% enters the tentative group.

2. Evaluate all 12 rules in priority order

01 Crowded area	no match
02 People + vehicles	no match
03 Signals	no match
04 Animals	no match
05 Tools	no match
06 Uncertain detections	**MATCH**
07 Dense objects	no match
08 Vehicles	no match
09 People nearby	no match
10 Empty area	no match
11 Sparse objects	no match
12 Moderate objects	no match

3. Choose primary plan

First matched action rule wins priority. All matching action plans and their steps are retained.

Uncertain-Detection

Take Action
tentative = 2 >= 1
Targets: d2 and d3

4. Recommend the step

Acquire another view before using either tentative label for an action.

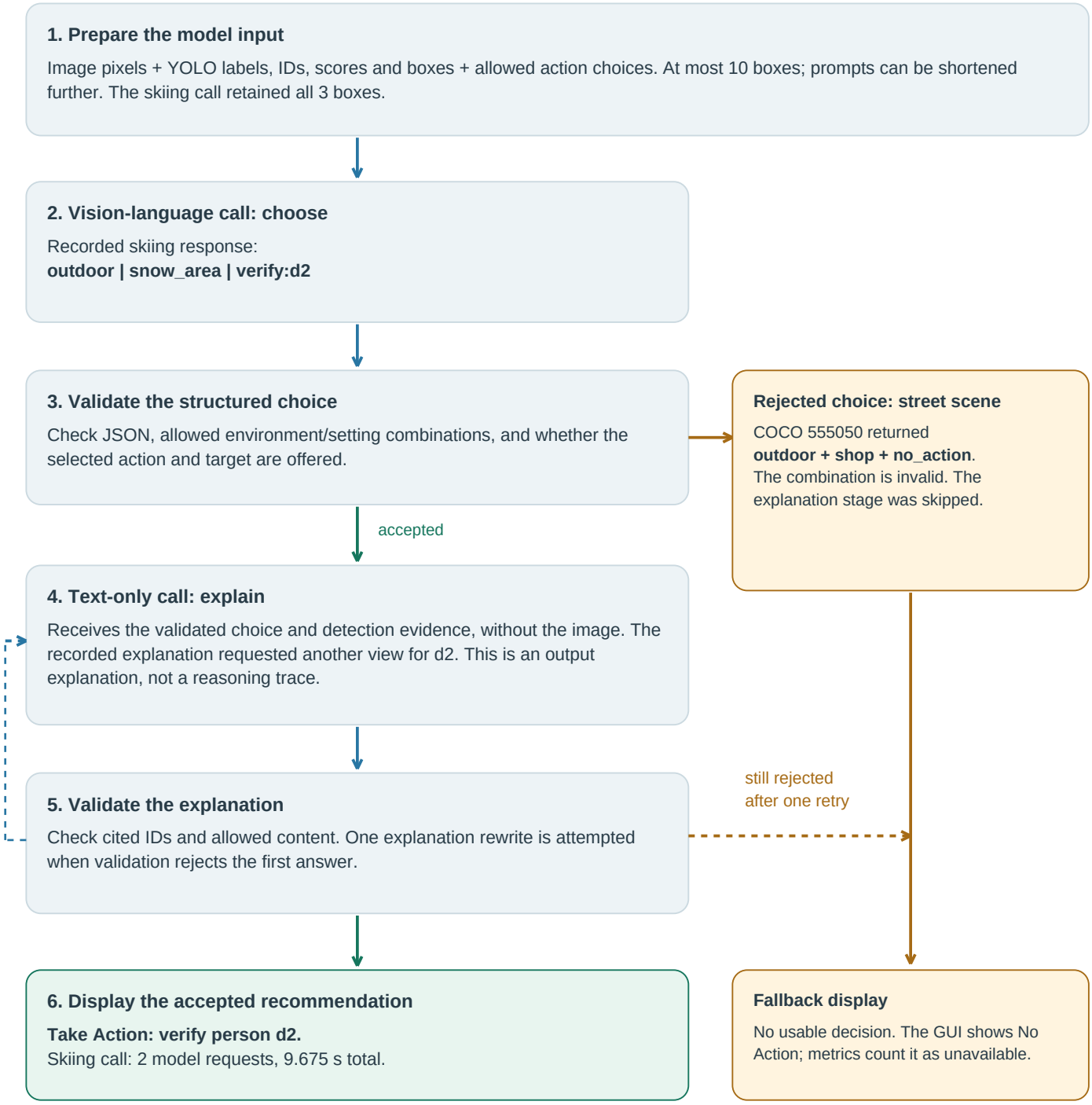
Why this differs from the annotation-based plan

The human annotations establish a person in this scene, so the robot reference selects **People-Presence**. The detector reports that person below 50%, so the measured algorithm selects **Uncertain-Detection**. Both request Take Action, but their primary plan names differ.

All 12 rules and 18 conditions were evaluated in the recorded call; this diagram does not imply that checking stops at rule 6. Idle plans apply only when no action rule matches. The application displays recommendations and does not execute physical robot commands.

LLM: propose, validate, explain

Observed application flow for Qwen3-VL 2B | Internal model reasoning is not recorded



What this diagram establishes

The green route is the saved skiing result, and the amber route is the saved street-scene rejection. The dashed retry route is implemented behavior and was not needed for the skiing example. With no eligible detections, the application bypasses the model and displays No Action.

The diagrams show prompts, responses and application checks. They do not claim access to the LLM's hidden reasoning. Validation acceptance does not establish that an explanation or action is correct.